

FALCON's Response to RFQ Delhivery for Robodome Sorter

March 23, 2024

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

Mr. Kaushal Agarwal

M/s Delhivery

Offer Ref: F24-00196; Date 21.03.2024.

Subject – Commercial Offer for Robodome Sorter.

We are pleased to submit our Offer to **Delhivery** in response to your **requirement for a Robodome Sorter**, we have taken this opportunity to also include adjustments discussed during various meetings.

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed Techno-commercial budgetary proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Adib Hasim

Sr. Manager – Sales & Solution

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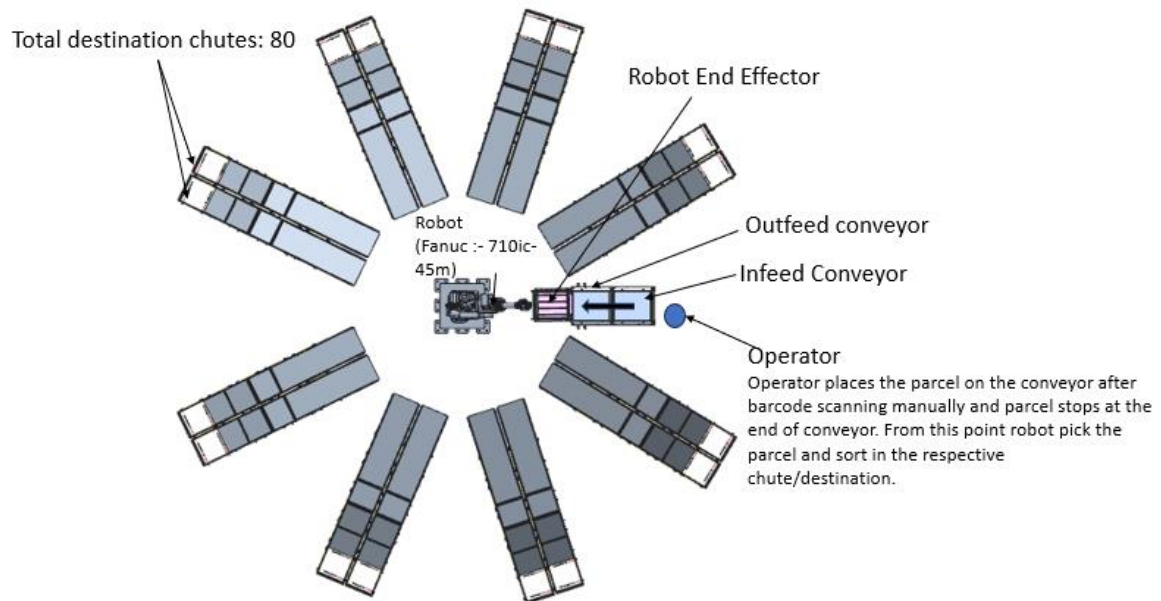
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1. Glossary

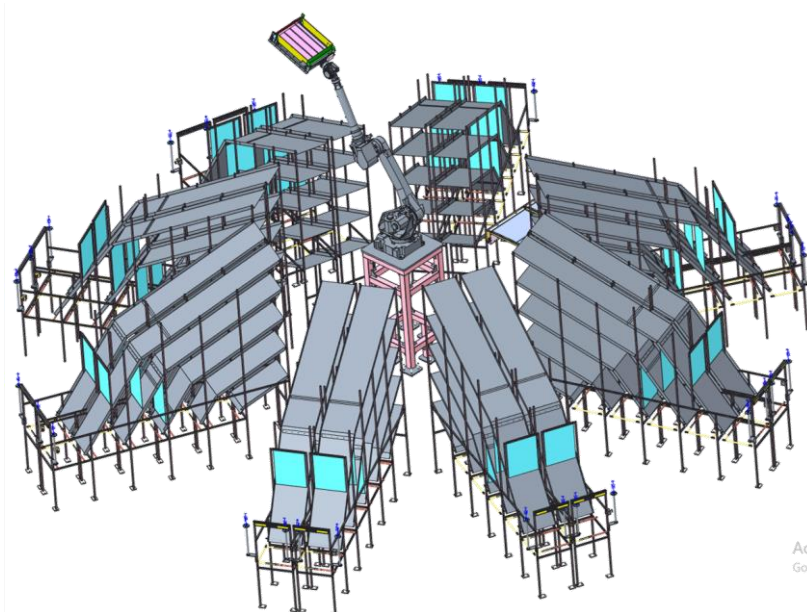
S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	AC	Alternating Current
4	DC	Direct Current
5	PLC	Programmable Logic Controller
6	BOQ	Bill Of Quantity
7	I/O	Input/ Output
8	PDP	Power Distribution Panel
9	PC	Personal Computer
10	UPS	Uninterrupted Power Supply

2. Detailed Solution Summary & Concept Layout

2.1 Layout View –



Plan View



Isometric View

2.2 Conveyable Product Specifications

Max Length	mm	500
Max Width	mm	400
Max Height	mm	300
Min Length	mm	100
Min Width	mm	50
Min Height	mm	10
Max Weight	Kg	6
Min Weight	Kg	0.05

3. Process Flow

The operators standing near the infeed conveyor pick the material from the trolleys and place them onto the infeed conveyor and do the scanning of Barcode by Manual Handheld Scanner.

Once the scanning will be done shipment will be transferred to the outfeed conveyor and robot end effector.

From the robot end effector, the parcels got transferred to the Robot's Carrier

Based on the scanning of the barcode, the material gets sorted and collected into the respective chute.

Once the chute/bag has been filled the operators will close the bag and replace them with new bag.

4. Proposed System Details

UAT Parameters	SLA
Barcode Reading	Manual Scanning through Handheld Scanner
Sorter TPH	~ 900 PPH for 80:20 Parcel Parity
Sorting Accuracy	99.90%

5. Proposed System Technical Details / BOM

5.1 Mechanical Equipment

Pos	Qty	Description	Value
FM1	1	Conveyor Package Infeed Conveyors	2.2 metres (2 Nos)
FM2	1	Chute Package Direct Bagging Chute with Bag Holders	80 Nos
FM 3	1	Robot Set	1 Set

5.2 Electrical Equipment

Electricals			
FC1	1	Electrical Cabling and Wiring Main Power Distribution Panel Main Control Panel Network Switches Field Cabling	Included

5.3 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls Barcode Scanning Controls Conveyor Controls PLC Based controller IT Control IT and Software Packages Networking Switches	Included

6. System Success Matrix/Handover to the Client

Falcon team will share the Success matrix parameters alongwith SOP. Basis on the completion of success matrix system will be handed over to the client .

7. System Warranty

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt from Various Maintenance Packages.

Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

8. Price Schedule

S. No	Items	Qty	UOM	Price
1	Conveyor Package	1	Set	₹ 2,41,570
2	Robodome Package	1	Set	₹ 31,53,846
3	Chute Package	1	Set	₹ 9,25,360
4	Field Wiring, Electrical Panel & Controls	1	Set	₹ 10,60,558
5	Installation & Commissioning	1	Set	₹ 2,69,067
6	Packing & forwarding	1	Set	₹ 1,61,440
7	Project Management & Engineering	1	Set	₹ 1,34,533
Total				₹ 59,46,374

9. Payment terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance.

Stage 2: 40% Against Delivery at Site with 100% Taxes

Stage 3: 10% Against Installation and Commissioning.

10. Terms & Conditions

Total Price is to be understood:

1. Ex-Works Greater Noida
2. Excluding Taxes & Duties
3. Excluding the Material Unloading and Laydown area at customer's site.

11. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

1. Server PC
2. Collection Trolleys.
3. Safety Netting, barricading, Bollards, toe guards or any such safety requirements shall be in Clients Scope.
4. Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
5. Electrical power supply and wiring to the main control cabinets.
6. UPS for Controls and Drives
7. Electrical Power for Installation
8. Network Cabling up to the Main Server Rack
9. Intermediate wiring to parts which are to be supplied by the Purchaser/others.
10. Emergency/Uninterruptable power supply
11. Fire-alarm and fire protection devices.
12. Traffic and route markings
13. Laydown Area
14. Cat walks, bridges, maintenance aisles and platforms
15. Assembly tools like forklifts and hoisting machines
16. All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
17. Any Kind of Civil work
18. Any adjustment of the Supplier's scope of supply to local rules and regulations
19. X-Ray machines
20. Roller cages/ Pallets
21. Simulation and 3D animation of the sorter system
22. Interface with other equipment not specified in this offer.
23. Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
24. The supply and installation of fencing around the different corridors.
25. Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.